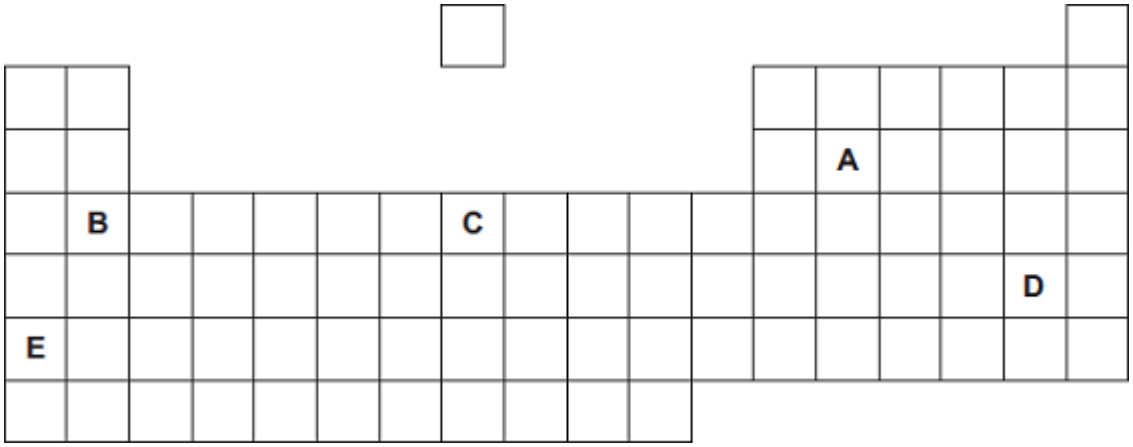
Q1.

The periodic table on the Data Sheet may help you to answer these questions.

Part of the periodic table is shown below.



The letters are **not** the symbols of these elements.

Choose your answers **only** from the letters shown in the periodic table above.

Which letter, **A**, **B**, **C**, **D** or **E**, represents:

(a) (i) an alkali metal Letter ☐

(1)

(ii) a transition element Letter ☐

(1)

(iii) a Group 4 element Letter ☐

(1)

(b) A chemistry teacher demonstrated the reaction between sodium and water to a class of students. One of the students wrote the following notes:

The reaction between sodium and water

A piece of sodium was cut easily into smaller pieces with a knife.

The sodium was added to some water in a trough.

The sodium:

- floated
- melted quickly to give a silvery ball
- moved on the surface of the water
- fizzed.

Use the information in the box to help you answer these questions.

What evidence is there that:

(i) sodium has a low melting point

_____

_____

(1)

(ii) sodium is soft

_____

_____

(1)

(iii) a gas was produced?

_____

_____

(1)

(Total 6 marks)

Q1.

(a) (i) E

1

(ii) C

1

(iii) A

1

(b) (i) quickly melted

allow melts in contact with water,
allow bp 100 °C (of water) shows mp is low
ignore one other piece of information

1

(ii) easily cut

ignore one other piece of information

1

(iii) effervescence / fizzing / bubbling

ignore named gas
ignore one other piece of information

1

[6]